

AK VisionFlow — Phase 5

Claude Development Prompt
Complete Accounting + Professional Procurement

Critical context: Phase 1–3 are already implemented, and Phase 4 is the production-hardening and Advanced Business Command Center phase. Phase 5 is an incremental upgrade to the existing repository. Inspect the current implementation first. Do not rebuild working modules, reset the database, or delete/corrupt existing data.

1. Mission

Transform AK VisionFlow from a strong operational POS/inventory platform into a proper business ERP by adding a robust accounting engine and a professional end-to-end procurement workflow. Accounting and procurement must integrate with the existing sales, purchases, inventory, payments, suppliers, customers, branches and dashboard systems.

2. Non-Negotiable Rules

- Preserve all existing Phase 1–4 functionality.
- Never reset, truncate or recreate the production database.
- All schema changes must use safe, versioned database migrations.
- Existing tenants, users, customers, suppliers, products, inventory, sales, purchases, invoices, payments, prescriptions and optical orders must remain intact.
- Maintain strict multi-tenant isolation on every new accounting and procurement record.
- Financial calculations must remain backend-authoritative.
- Use transactional database operations for financial posting and stock-affecting workflows.
- Financial records must be auditable; avoid destructive deletion of posted transactions.
- Respect existing RBAC and branch permissions.
- Do not hard-code Khalid Eye Clinic or any single customer; the product remains generic SaaS.
- Run existing automated tests before and after changes and add comprehensive regression tests.

3. Accounting Architecture

Implement a proper double-entry accounting subsystem integrated with the existing business transactions. Do not create a disconnected reporting-only accounting module. Sales, purchases, expenses, payments, refunds/reversals and other relevant transactions should produce correct accounting effects according to configured accounts and business rules.

Core accounting components:

- Chart of Accounts with account types, groups, codes, parent/child hierarchy and active/inactive status.
- Double-entry Journal Entries with balanced debit/credit lines.
- General Ledger with date, account, branch, reference and transaction drill-down.
- Journal Entry source/reference linking back to the originating business transaction.
- Cash accounts and Cash Book.
- Bank accounts and Bank Book.
- Customer Accounts Receivable.
- Supplier Accounts Payable.
- Customer Ledger and Supplier Ledger.

- Payment vouchers and Receipt vouchers.
- Opening balances with controlled setup workflow.
- Accounting periods / financial years.
- Period locking/closing so posted historical transactions cannot be silently changed.
- Reversal/correction workflow instead of destructive deletion.
- Accounting audit trail.

4. Financial Reports

- Trial Balance
- Profit & Loss Statement
- Balance Sheet
- Cash Flow Statement
- General Ledger
- Cash Book
- Bank Book
- Accounts Receivable Aging
- Accounts Payable Aging
- Customer Ledger
- Supplier Ledger
- Expense Summary
- Income Summary
- Sales vs Cost vs Gross Profit
- Branch-wise Profit & Loss
- Account-wise transaction report

5. Accounting Integration Rules

- Sale → revenue, receivable/cash/bank and applicable tax/discount effects.
- Purchase/GRN/invoice → inventory/cost and payable effects according to the selected workflow.
- Customer payment → receivable reduction and cash/bank increase.
- Supplier payment → payable reduction and cash/bank reduction.
- Expense → expense account and cash/bank/payable effect.
- Refund/reversal → correct reversal of the original accounting effect; no phantom customer balance.
- Stock adjustments with financial impact must follow controlled accounting rules where applicable.
- All postings must be traceable from report → journal → source transaction.

6. Professional Procurement Workflow

Implement procurement as a controlled lifecycle rather than a simple purchase-entry screen:

Purchase Request → RFQ → Supplier Quotations → Quotation Comparison → Approval → Purchase Order → Goods Receipt/GRN → Inventory Update → Supplier Invoice → Payment → Supplier Ledger

- Purchase Requisition / Purchase Request.

- RFQ creation and supplier selection.
- Supplier quotation entry, including item, quantity, price, tax, discount, validity and delivery terms.
- Side-by-side quotation comparison.
- Best-quotation recommendation based on configurable criteria.
- Purchase approval workflow with configurable thresholds.
- Purchase Orders with status lifecycle.
- Partial PO receiving.
- Goods Received Note (GRN).
- Rejected/damaged/short received quantities.
- Automatic inventory update after approved receiving.
- Supplier invoice recording.
- Supplier payment allocation.
- Purchase return workflow.
- Full procurement history and audit trail.

7. Supplier Management

- Supplier profile and contacts.
- Supplier-specific products and purchase prices.
- Purchase history.
- Quotation history.
- PO history.
- GRN history.
- Supplier invoice history.
- Outstanding payable balance.
- Payment history.
- Supplier performance metrics.
- Average delivery time.
- Price trend/history.
- Supplier-wise profitability/cost analysis where meaningful.

8. Inventory & Procurement Integration

- GRN must update stock only after the appropriate approval/receipt state.
- Maintain product cost history.
- Support partial receiving and remaining quantities.
- Track stock by branch/warehouse where supported by existing architecture.
- Preserve inventory valuation consistency.
- Do not allow duplicate receipt/sync operations to inflate stock.
- Purchase returns must correctly reverse stock and financial effects.
- Procurement data should become available to the Advanced Business Command Center.

9. Tax & Pakistan Readiness

- Design tax configuration so rates are not hard-coded.
- Support applicable sales tax/GST concepts.
- Support tax-inclusive and tax-exclusive pricing where appropriate.
- Track tax components in accounting and reports.
- Provide a foundation for NTN/STRN and FBR/e-invoicing integration where applicable.
- Keep the tax engine configurable because legal/tax requirements can change.

10. Advanced Command Center Integration

Extend the Phase 4 Advanced Admin/Owner Business Command Center with accounting and procurement intelligence.

- Revenue, gross profit and net profit from accounting-authoritative data.
- Receivables and overdue receivables.
- Payables and overdue payables.
- Cash and bank position.
- Purchases and procurement pipeline.
- Pending purchase approvals.
- Pending POs.
- Pending GRNs.
- Supplier payment commitments.
- Purchase price changes.
- Branch-wise financial performance.
- Drill-down from every KPI to underlying records.

11. UX Requirements

- Keep workflows simple enough for small and medium optical/medical businesses.
- Use clear statuses and progress indicators for procurement.
- Provide searchable/filterable tables and useful date/branch/supplier filters.
- Show validation errors clearly.
- Prevent accidental posting of incomplete or unbalanced accounting entries.
- Use confirmation steps for irreversible/financial actions.
- Maintain existing responsive/PWA/light-dark UI patterns.
- Do not overload ordinary shop users with accounting controls they are not authorized to access.

12. RBAC & Approvals

- Only authorized roles can create/post/approve journal entries.
- Only authorized roles can approve POs and procurement requests.
- Define permission checks at API/backend level, not only in the UI.
- Support approval thresholds for purchases, expenses and other configurable actions.
- Record who created, approved, posted, received, invoiced and paid each procurement transaction.

13. Testing Requirements

- Accounting debits and credits must always balance.
- Test sales posting, purchase posting, payments, receipts, refunds and reversals.
- Test customer and supplier balances against journal activity.
- Test P&L, Balance Sheet and Trial Balance calculations using controlled fixtures.
- Test procurement lifecycle from request through payment.
- Test partial receiving, short/damaged receiving and purchase returns.
- Test tenant isolation and branch permissions.
- Test duplicate prevention and idempotency for financial and stock operations.
- Test migration on a copy of realistic existing data.
- Run the complete existing regression suite after implementation.

14. Migration & Data Safety

Before changing the schema, inspect the existing Prisma/PostgreSQL models and migrations. Design accounting/procurement tables so they integrate with existing IDs and records rather than duplicating core entities. Where historical transactions cannot be fully back-posted automatically, create a controlled opening-balance or migration strategy and document it. Never silently fabricate financial history.

15. Required Deliverables

- Complete accounting module and database schema.
- Chart of Accounts management.
- Journal/ledger engine.
- Financial reports.
- Customer/Supplier receivable/payable integration.
- Professional procurement lifecycle.
- Quotation comparison.
- Purchase approvals.
- PO and GRN workflow.
- Supplier invoice and payment workflow.
- Tax-ready accounting structure.
- Command Center financial/procurement widgets.
- Database migrations.
- Automated tests and regression results.
- Updated technical documentation.
- Phase 5 completion report with known limitations and next recommendations.

16. Acceptance Criteria

- Existing Phase 1–4 functionality continues to work.
- No existing customer/tenant data is deleted or corrupted.
- Accounting entries are balanced and auditable.
- Sales, purchases, payments, receipts, refunds and reversals integrate correctly with accounting.

- Customer and supplier balances reconcile with accounting records.
- Procurement supports the full Request/RFQ/Quotation/Approval/PO/GRN/Invoice/Payment lifecycle.
- Inventory is updated correctly by approved receiving and returns.
- All sensitive accounting/procurement actions respect RBAC and tenant isolation.
- Financial and stock operations are transaction-safe and idempotent.
- Advanced Command Center displays accurate financial/procurement information.
- All existing tests pass and new Phase 5 tests pass.
- The system is ready for the next phase: Multi-Branch + Warehouse + Advanced Business Control.

17. Claude Execution Order

1. Inspect the current repository, Prisma schema, migrations, APIs and existing Phase 1–4 workflows.
2. Run the current test suite and establish a baseline.
3. Map existing sales/purchase/payment/inventory entities to the accounting design.
4. Design accounting and procurement schema without duplicating existing core entities.
5. Implement migrations and test them against realistic existing data.
6. Implement Chart of Accounts and accounting engine.
7. Implement journals, ledgers and financial reports.
8. Integrate existing transactions with accounting.
9. Implement procurement lifecycle and approvals.
10. Implement GRN/inventory integration and supplier invoicing/payment.
11. Integrate new financial/procurement metrics into the Command Center.
12. Add comprehensive automated/regression tests.
13. Run the complete suite and perform data-integrity/security review.
14. Document implementation, migrations, limitations and operational procedures.

Final Instruction to Claude

Build Phase 5 as a safe, production-quality incremental upgrade. Do not rebuild the existing core and do not reset the database. The accounting engine must be a real double-entry system, not a collection of summary reports. Procurement must be a complete controlled workflow, not merely a purchase form. Integrate both with existing sales, inventory, suppliers, customers, payments, branches and the Advanced Command Center. Preserve tenant isolation, data integrity, offline/sync safety and backward compatibility. Every important financial and stock operation must be auditable, transaction-safe and covered by automated tests.